

PhytoGraph — cycles 7-9

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Contents

PhytoGraph — cycles 7-9	1
Abstract	1
Introduction	2
Approach	2
Cycle 7: Barrier 1 Repair	2
Cycle 8: Wave 2 Track Enrichment	3
Cycle 9: Barrier 2 Integration	11
Findings	12
Discussion	12
Open Questions	12
References	13
Appendix: Implementation Details	13

PhytoGraph — cycles 7-9

Abstract

Cycles 7-9 moved PhytoGraph from a repaired shared substrate into Barrier 2-ready Wave 2 enrichment.

Cycle 7 repaired and validated the Barrier 1 canonical-member projection defect that had blocked downstream work. The repaired substrate now propagates synonym-resolved accepted keys into retained hyperedges and deduplicates on full typed member sets rather than raw taxon names. The substrate validator passed with 363,237 nodes and 641,183 retained hyperedges.

Cycle 8 launched Wave 2 track enrichment across all six tracks. Each branch wrote only to its own `tracks/trackN/` namespace, preserved the frozen substrate as read-only, and kept enrichment distinct from prediction. The branches produced per-track `ENRICHMENT_AUDIT.md` files and machine-readable enrichment outputs.

Cycle 9 integrated the six Wave 2 branches as Barrier 2 candidate inputs. The conformance validator passed for all six tracks, the combined track regression suite passed with 41 tests, `prediction_ledger.tsv` remained empty, and the integration report preserved each branch’s evidence boundaries and data-limited caveats.

The result is not a predictive-instrument milestone. No Track 1 reticulation index, Track 2 ranker, Track 3 convergence-pressure statistic, Track 4 crop-substitution engine, Track 5 chemodiversity predictor, or Track 6 live model probe was built. The campaign is now positioned for Barrier 2 review and later Wave 3 instrument construction.

Introduction

The PhytoGraph campaign depends on a shared typed hypergraph substrate before any track-specific predictive instruments can be trusted. Earlier cycles created Wave 1 source staging and attempted a Barrier 1 substrate freeze, but that freeze failed audit because canonical keys were not propagated back into retained hyperedges and deduplication collapsed distinct biological assertions.

Cycles 7-9 address the transition from substrate repair to track enrichment:

- **Cycle 7** repaired Barrier 1 canonical-member projection and deduplication.
- **Cycle 8** launched six independent Wave 2 enrichment branches.
- **Cycle 9** integrated those branches into a Barrier 2 conformance package.

Key terms:

- **Barrier 1**: the point where staged source rows become the shared read-only substrate.
- **Barrier 2**: the point where all track-specific enrichment outputs are checked for schema conformance, provenance discipline, duplicate-edge issues, and prediction-boundary discipline.
- **Enrichment**: observed or cited evidence attached to the frozen substrate. Enrichment is not prediction.
- **Prediction ledger**: the master ledger where falsifiable model-generated predictions will later be filed. It stayed empty in these cycles.

Approach

The cycles followed the campaign's dependency discipline. Cycle 7 kept Wave 1 source rows fixed and repaired only the Barrier 1 substrate pipeline. Cycle 8 then treated the repaired substrate as read-only and launched six track-local enrichment branches. Cycle 9 did not build instruments; it validated that branch outputs were suitable candidate inputs for Barrier 2.

The integration checks used:

```
python3 tools/validate_barrier2_track_enrichment.py
python3 -m pytest -q tests/test_track1_reticulation_enrichment.py tracks/track2/tests/test_ghost_enrichment.py
↳ tracks/track3/tests/test_track3_enrichment.py tracks/track4/tests/test_domestication_enrichment.py
↳ tracks/track5/tests/test_track5_enrichment.py tests/test_track6_offline_probe.py
python3 tools/validate_barrier1_substrate.py
python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace
python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace
```

The reported results were:

```
PASS: Barrier 2 track enrichment conformance (6 tracks checked)
41 passed in 20.87s
PASS: Barrier 1 substrate validation (363237 nodes, 641183 retained hyperedges)
```

Cycle 7: Barrier 1 Repair

Cycle 7 began with researcher session 00ecd932-4314-43ed-8a94-b64661660769, which scoped the remaining Barrier 1 blocker as canonical-member projection loss. The prior broken pipeline had deduplicated large source frames using raw-name-only member sets, so same-taxon rows with different traits or compounds could collapse.

Worker session e63dca56-ad6d-4877-9e8c-6cf590e38f5e repaired:

- scripts/barrier1_common.py
- scripts/barrier1_merge_substrate.py
- scripts/barrier1_apply_synonyms.py
- scripts/barrier1_deduplicate_edges.py
- scripts/barrier1_write_reports.py
- tools/validate_barrier1_substrate.py
- tests/test_barrier1_canonical_members.py
- tests/test_barrier1_merge.py

The worker produced:

- substrate/BARRIER1_REPAIR_REPORT.md
- phytograph_dataset/canonical_member_audit.tsv
- phytograph_dataset/resolved_key_propagation_audit.tsv
- phytograph_dataset/dedup_collision_audit.tsv
- phytograph_dataset/dedup_before_after_by_edge_type.tsv
- substrate/barrier1_dedup_before_after.png
- substrate/barrier1_canonical_member_widths.png
- reports/barrier1_canonical_member_repair_validation.md

The repaired validator passed:

```
PASS: Barrier 1 substrate validation (363237 nodes, 641183 retained hyperedges)
10 passed in 31.86s
```

Auditor session 67171417-7a93-4bef-bcf8-16b65a264c6d validated the repair. The decisive probes were:

Probe	Result
Track 3 resolved retained rows with blank accepted key	0
Track 3 resolved retained rows still pending crosswalk	0
Track 3 rows missing accepted key in canonical members	0
Track 3 width-one canonical member sets	0
Track 5 resolved retained rows with blank accepted key	0
Track 5 resolved retained rows still pending crosswalk	0
Track 5 rows missing accepted key in canonical members	0
Track 5 width-one canonical member sets	0
Acaena x ovina retained Track 3 rows	14
Acaena x ovina unique canonical member sets	14

The repair also corrected the Tier 0 reporting distinction: accepted taxonomy rows are 60,000, while 113,582 synonym rows are separate synonym coverage, not accepted taxa.

Cycle 8: Wave 2 Track Enrichment

Cycle 8 researcher session 3edad7dc-2222-4bec-8bc0-2739c0c80d5b launched the first post-Barrier-1 fan-out. The shared rules were: read phytograph_dataset/ as frozen, write only under each track namespace, do not change schema v1.0, do not independently normalize synonyms, do not emit predictive instruments, and preserve provenance, license, caveats, and evidence scope.

Track 1 Reticulation

Track 1 projected the 28-row M1.3 reticulation seed set onto the repaired Barrier 1 accepted-key namespace. It remained data-limited.

Edge type	Staged rows	Resolved	Pending
chromosome_count_assertion	12	2	10
ploidy_state_assertion	6	1	5
hybridization_event	1	0	1
polyploidization_event	4	0	4
reticulate_inheritance_evidence	5	0	5
Total	28	3	25

The audit explicitly forbids using these rows to compute a `tree_compatibility_index` or a planet-scale reticulation index. The canonical validation seeds for Track 1 remain mostly pending_crosswalk.

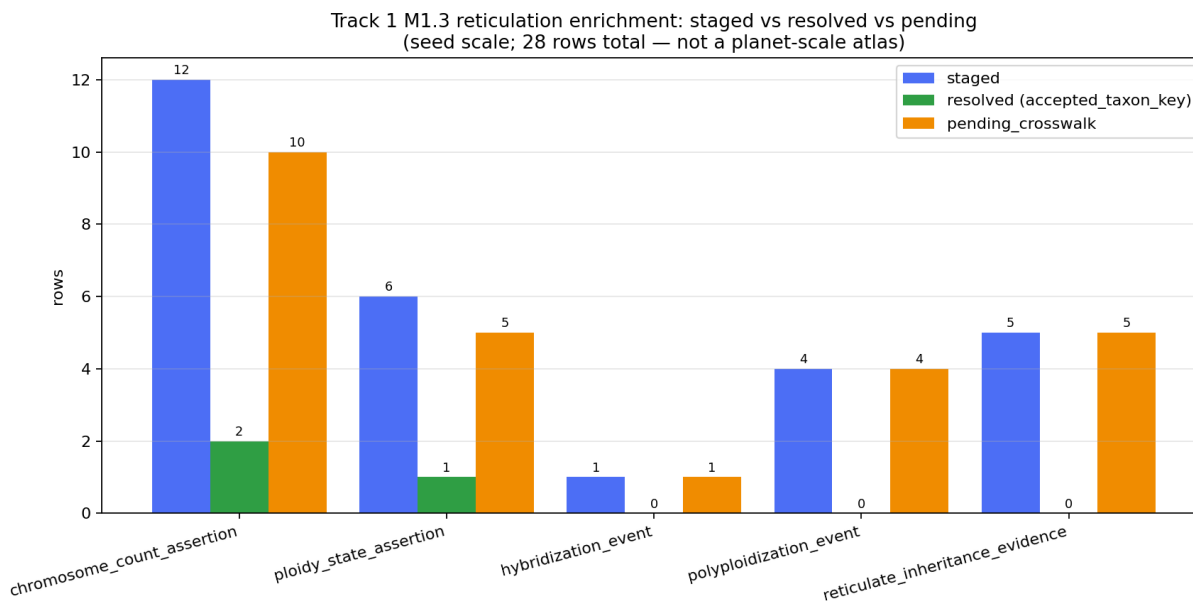


Figure 1: Track 1 reticulation enrichment coverage by edge type.

Track 2 Ghost Partners

Track 2 converted cited paleobotany and Janzen-Martin candidate rows into Track 2-local seed evidence. It did not infer anachronism predictions.

Metric	Value
Candidate seed edges	31
Candidate classes	6
Unique plant names	24
Unique extinct-fauna nodes in seed edges	11
Pending-crosswalk seed rows	25

Metric	Value
Resolved seed rows	6
Range-context support edges	52
Extinct-fauna support nodes	237
Paleo-context support nodes	80
Modern-disperser context nodes	49
Inferred candidate rows emitted	0
Prediction-ledger rows written	0

Every seed row was marked `literature_curated_seed`, `prediction_status=not_prediction`, `enters_prediction_ledger=false`, and `inferred_anachronism_claim=false`.

Track 3 Convergence

Track 3 projected the repaired substrate's AusTraits-derived trait, fruit morphology, and life-form edges onto a canonical Track 3 trait axis. It emitted `track3_trait_membership` rows only and no `convergence_signature` rows.

The output contained 209,297 memberships. Coverage by canonical trait included 12 nonzero traits and three zero-data traits (`ant_domatia`, `carnivory`, `parasitism`). Only `fleshy_fruit` and `capsule` cleared the 500 accepted-taxon floor.

Trait	Retained edges	Accepted taxa	Pending-crosswalk taxa	Floor met
<code>fleshy_fruit</code>	4,748	716	3,371	yes
<code>capsule</code>	9,074	543	8,531	yes
<code>c4_</code>	1,378	157	1,219	no
<code>photosynthesis</code>				
<code>myrmecochory</code>	2,195	288	849	no
<code>_other</code>	184,218	3,252	29,285	diagnostic bucket

The `_other` bucket was the main integration divergence. It is large because AusTraits includes many labels outside the Track 3 canonical axis. The cycle 9 integration fixed the validator interpretation: `_other` is recorded as a Barrier 2 review item, not treated as a canonical trait floor.

Track 4 Domestication

Track 4 attached M1.6 domestication staging rows to accepted substrate keys at data-limited scale. It retained six observed hyperedges: two `crop_pedigree`, two `vavilov_center_hyperedge`, and two `cultivation_or_domestication`.

Category	Staged rows	Joined rows	Unjoined rows	Main shortfall
<code>crop_pedigree</code>	43	2	41	focal accepted-key gaps

Category	Staged rows	Joined rows	Unjoined rows	Main shortfall
vavilov_center_hyperedge	43	2	41	focal accepted-key gaps
cultivation_or_domestication	104	2	102	focal accepted-key gaps
crop_wild_relative_pairs	69	3	66	crop and/or wild ancestor accepted-key gaps
heldout_validation_seed	22	2	20	held-out focal accepted-key gaps
climate_envelope	375	36	339	all rows data-limited until bioclim extraction

The held-out validation seed retained zero species-level overlap with retained crop-pedigree training evidence. Climate rows remained placeholders; no row supported climate suitability prediction.

Track 5 Chemodiversity

Track 5 projected resolved phytochemical and ethnobotanical evidence into its namespace, preserved a compound-level bioactivity firewall, and foregrounded source-density diagnostics.

Key outputs:

Artifact	Rows	Purpose
track5_enrichment_edges.parquet	23,524	Resolved taxon-keyed phyto and ethnobotany projection
track5_bioactivity_assertions.parquet	28,733	Compound-level bioactivity, no taxon column
track5_compound_class_membership.parquet	9,500	Duke CHEMCLASS mapping
family_chemistry_coverage_summary.tsv	91 families	Per-family coverage
per_taxon_screening_intensity.tsv	1,258 taxa	Source/compound/paper intensity

The dominant finding is source concentration. Dr. Duke accounted for 99.96% of the combined enrichment plus bioactivity signal. Removing Dr. Duke eliminated all family-cells above the 100-assertion floor, all bioactivity-class signal, and 99.2% of taxa with assertions.

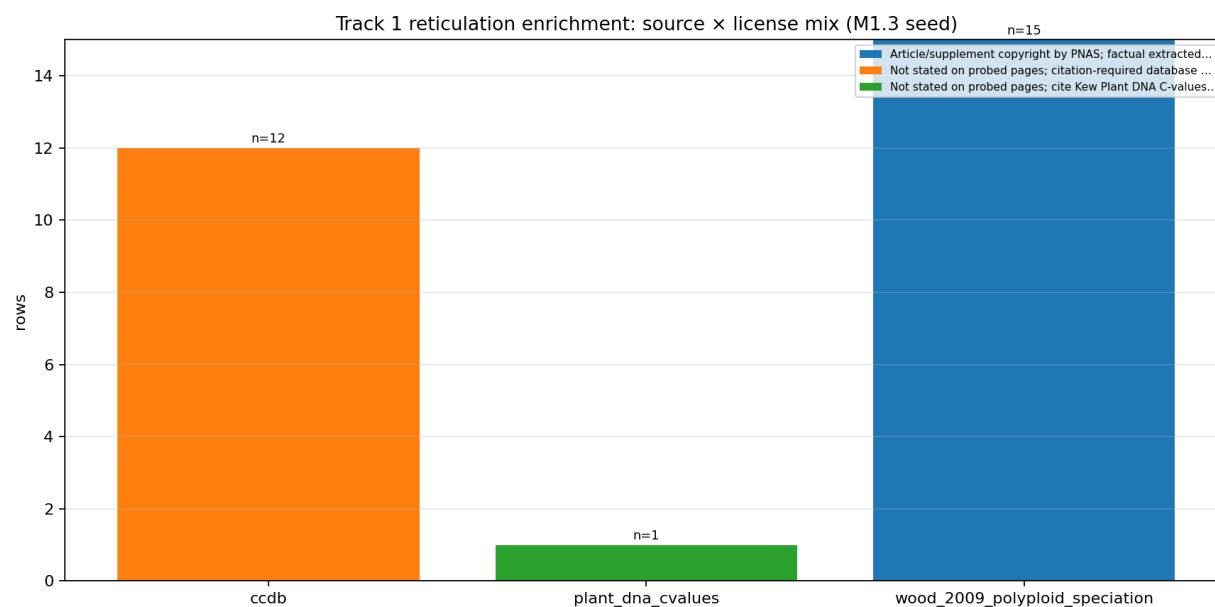


Figure 2: Track 1 reticulation source and license mix.

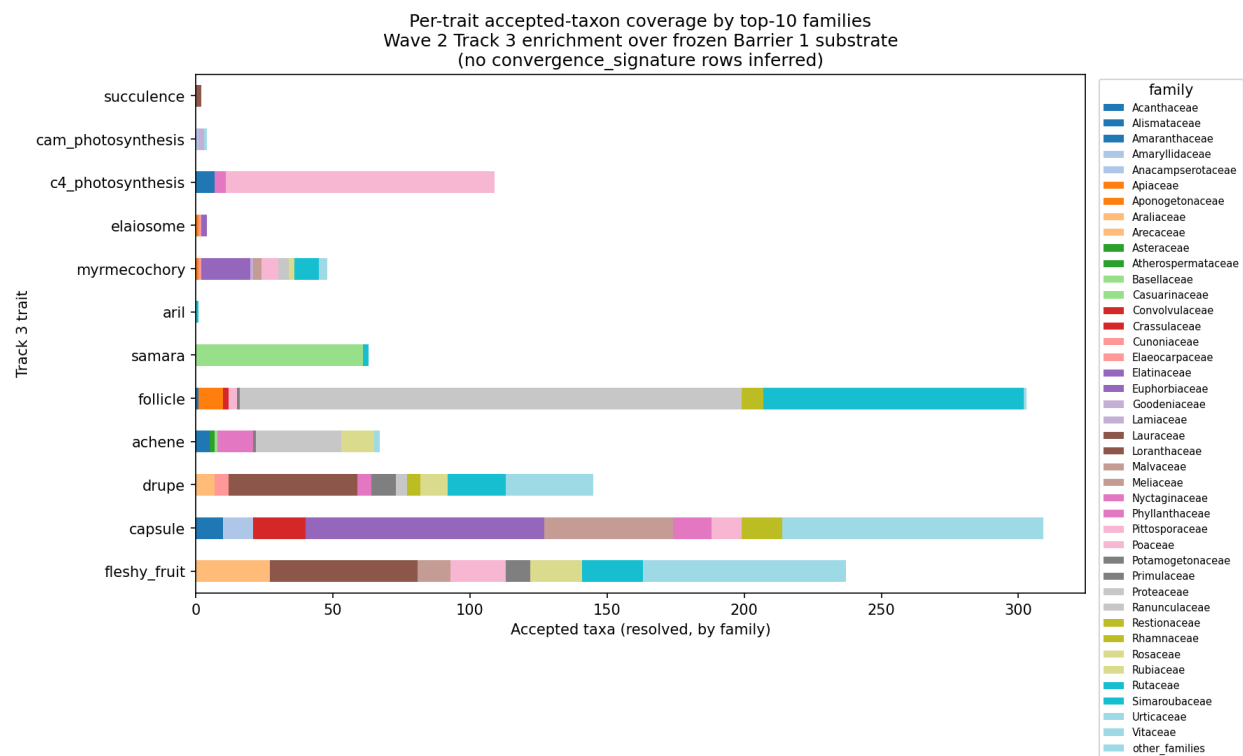


Figure 3: Track 3 trait coverage by family.

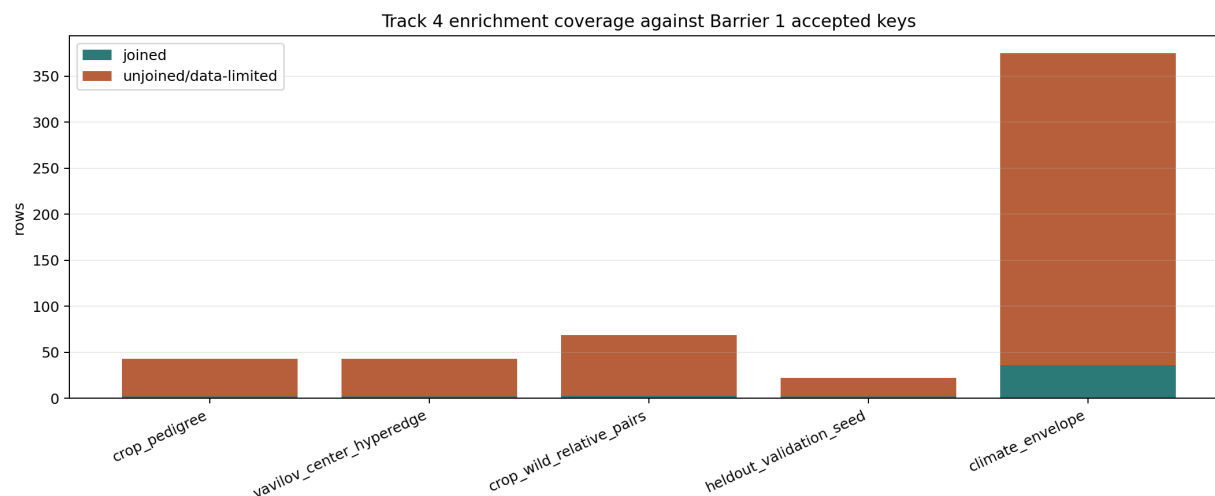


Figure 4: Track 4 joined and unjoined enrichment counts by evidence category.

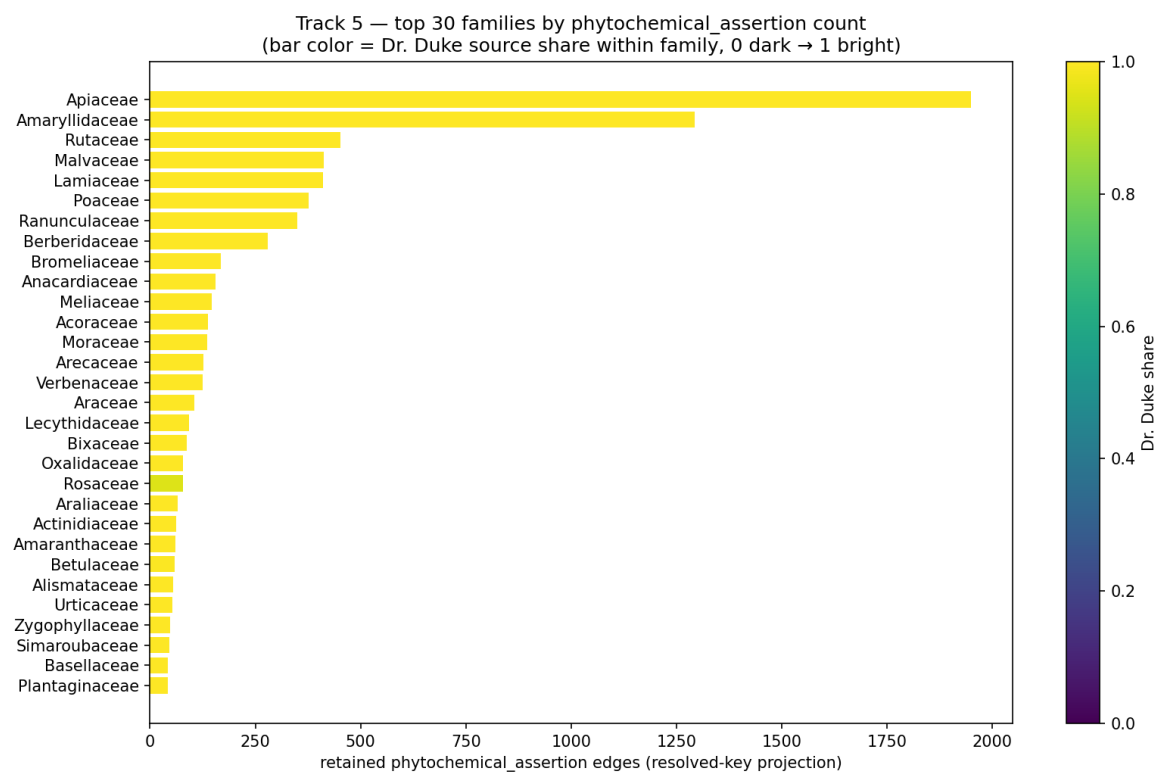


Figure 5: Top-30 Track 5 families by phytochemical assertion count, colored by Dr. Duke share.

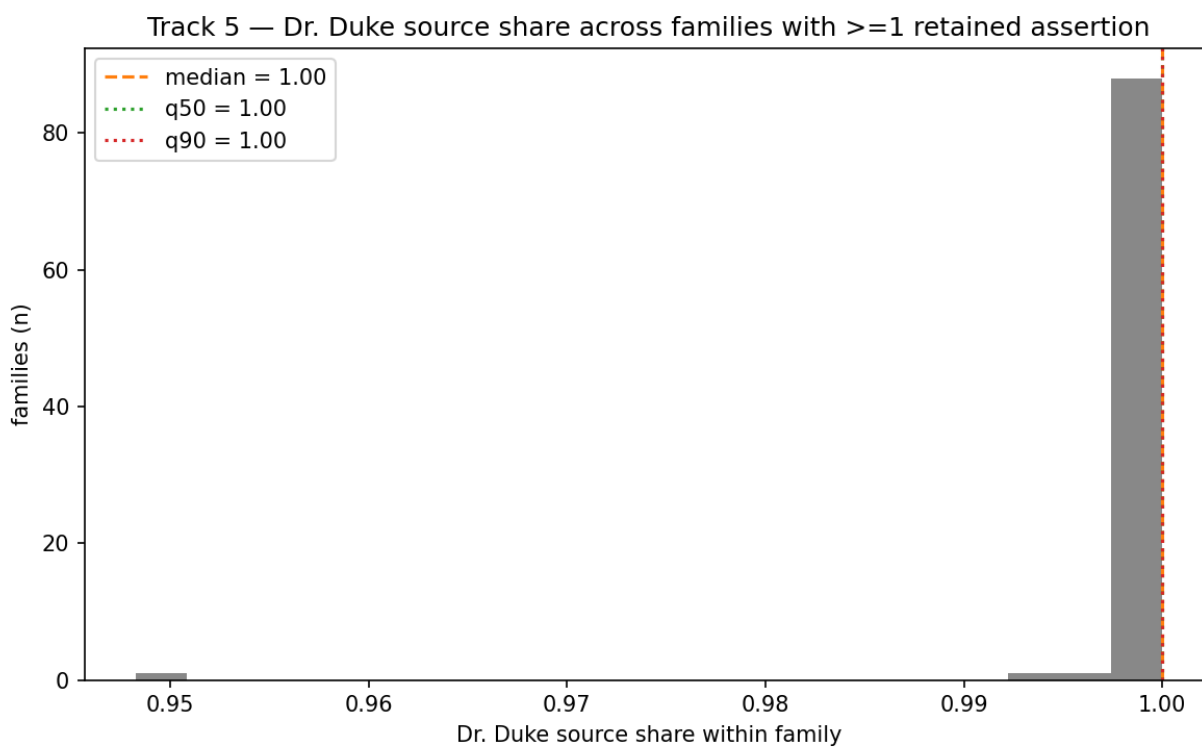


Figure 6: Distribution of Dr. Duke source share across retained Track 5 families.

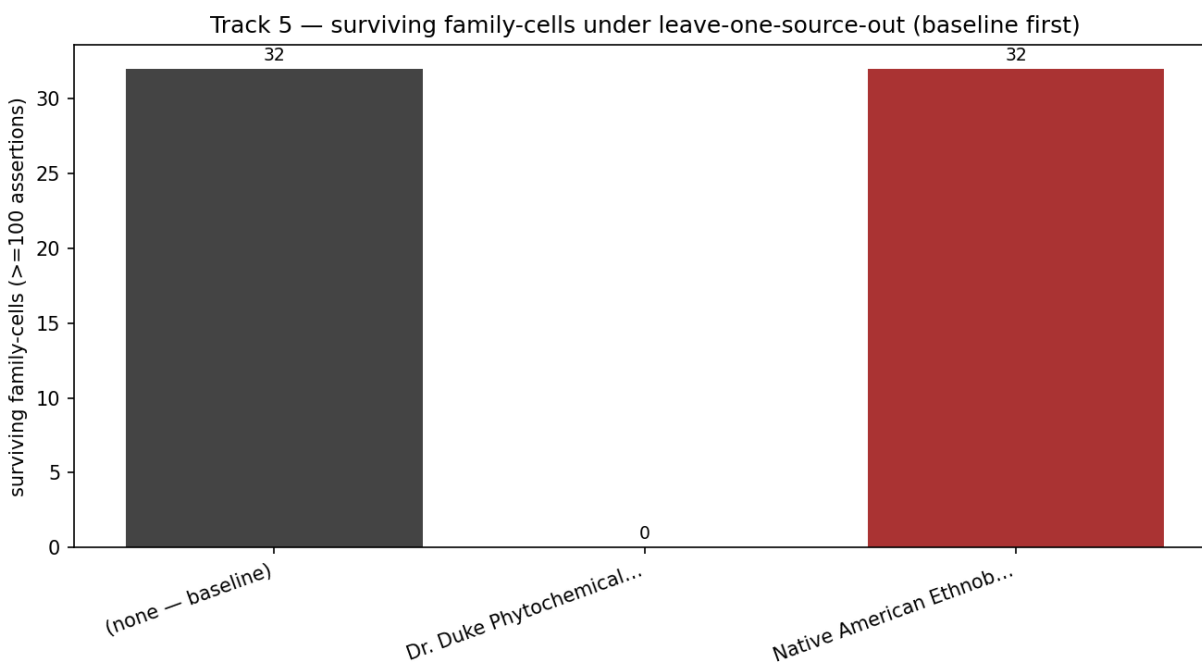


Figure 7: Track 5 leave-one-source-out coverage; Duke removal collapses all floor-clearing family-cells.

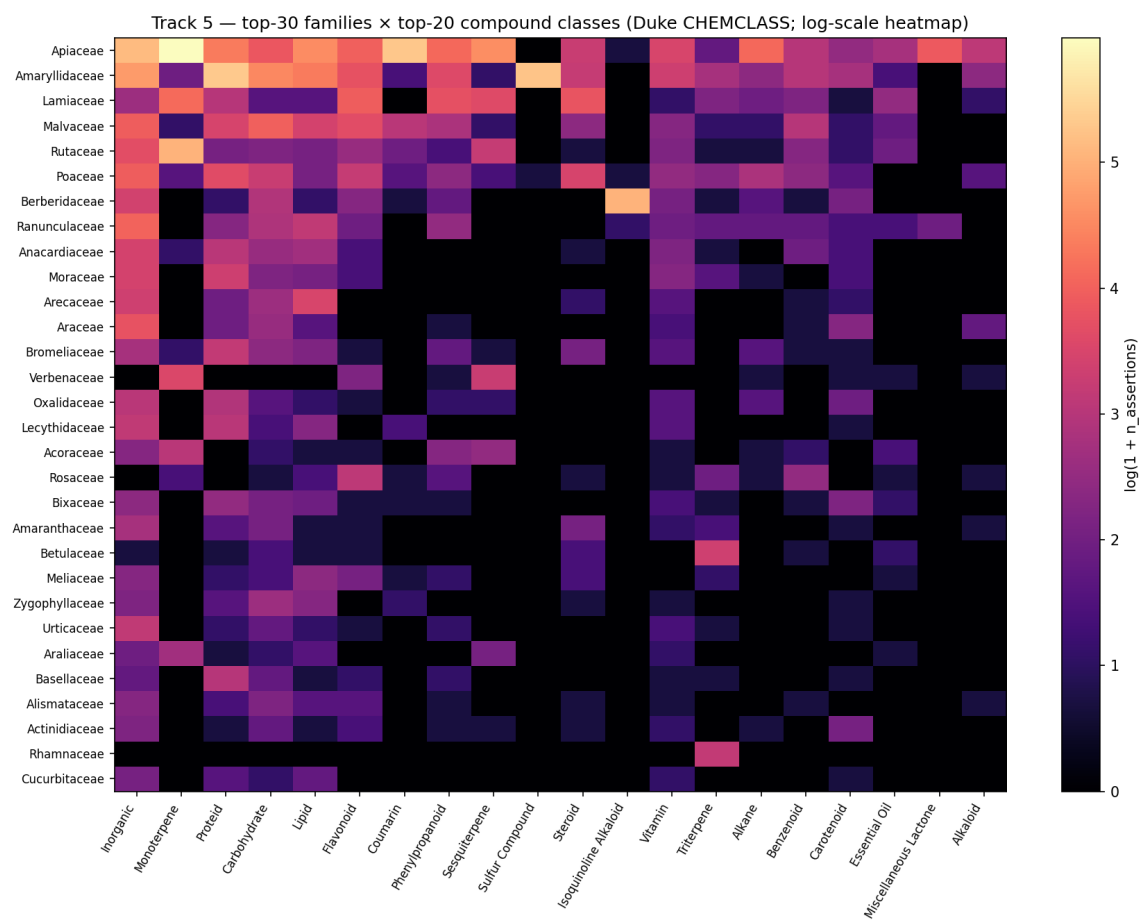


Figure 8: Track 5 family by compound-class matrix.

Track 5 did not emit chemodiversity_signature rows and did not convert ethnobotanical use into clinical bioactivity.

Track 6 Offline Probe

Track 6 built a static, free/open-source/offline probe question bank and ground-truth edge layer. It did not call or configure paid/key-gated providers and did not use the superseded M1.8 live-provider harness.

Category	Questions
convergence_detection	30
ghost_partner_reasoning	30
hybrid_pedigree	30
phytochemistry_safety	30
region_conditional	30
synonym_confusion	30
toxicity_lookalike_media_scope	30
Total	210

The branch also produced 210 schema-shaped adversarial_probe_edge rows. Every row uses foundation_model_response:offline_unrun, which is a placeholder, not a model output. Live or local-open model scoring remains future M3.T6 work.

Cycle 9: Barrier 2 Integration

Cycle 9 worker session 7e8a8f0a-786b-484b-8c69-d41b89087cc2 integrated the six Wave 2 branches and produced:

- tools/validate_barrier2_track_enrichment.py
- data/barrier2_track_enrichment_conformance.json
- reports/barrier2_wave2_integration_report.md

A ledger event was appended: b0c3ef94-5b71-4b59-9e66-6b50514386e4.

The integration report states that all six branches are candidate Barrier 2 inputs, not predictive instruments. prediction_ledger.tsv remains empty.

Track	Barrier 2 integration status	Integrated finding
Track 1	ready_data_limited	28 reticulation seed rows retained; 25 remain pending crosswalk
Track 2	ready_seed_scale	31 cited seeds and 52 range-context supports remain not-prediction
Track 3	ready_with_conformance_review_item	209,297 trait memberships retained; _other bucket requires explicit interpretation

Track	Barrier 2 integration status	Integrated finding
Track 4	ready_data_limited	6 observed domestication edges retained; CWR and bioclim evidence remain data-limited
Track 5	ready_with_source_bias_warning	23,524 resolved taxon-keyed rows retained; Duke share is 0.999598
Track 6	ready_offline_static	210 offline questions and 210 ground-truth rows retained; no live responses

Findings

The core substrate risk from earlier cycles is resolved. Barrier 1 now passes validation with full canonical-member projection and retained accepted-key propagation.

Wave 2 enrichment is successful as enrichment, not as prediction. Each track produced evidence-bearing artifacts in its own namespace, and the integration validator confirmed six-track conformance.

The strongest cross-track constraint is data limitation. Track 1 is seed-scale; Track 4 has no observed bioclim vectors; Track 5 is dominated by Dr. Duke; Track 6 is offline-static only; Track 3 has a large `_other` diagnostic bucket; Track 2 remains literature-seed scale.

The master prediction boundary held. No branch wrote predictive instrument outputs, and `prediction_ledger.tsv` remained empty.

Discussion

Cycles 7-9 are best understood as infrastructure-to-enrichment transition cycles. They did not produce falsifiable biological predictions yet. Their contribution is that the campaign now has a validated substrate and six track-local enrichment layers with explicit caveats.

The most important design discipline preserved across the work is separation of evidence from prediction. Track 2 seed hypotheses are not inferred ghost-partner predictions. Track 3 trait memberships are not convergence signatures. Track 4 pedigree and Vavilov edges are not crop-substitution recommendations. Track 5 compound bioactivity is not taxon-level clinical efficacy. Track 6 offline-unrun edges are not model responses.

The next campaign step should either move into Barrier 2 schema/provenance review or start carefully bounded Wave 3 instrument design with the recorded data-limited constraints carried forward.

Open Questions

1. Should Track 1 run a side-wave accepted-key recovery for canonical polyploid validation seeds before any reticulation index is attempted?
2. Should Track 3 treat `_other` as a diagnostic-only bucket permanently, or should a side-wave refine noncanonical `AusTraits` labels into additional instrument axes?
3. Can Track 4 recover enough CWR and bioclim evidence to support a Crop Substitution Engine, or must M3.T4 remain a design-only artifact?

4. Can Track 5 obtain non-Duke counterfactual coverage from NAEB synonym recovery, NPASS, KNApSAcK, or ChEBI before prediction?
5. When Track 6 moves beyond static construction, which free/open/local model path satisfies the no paid/key-gated constraint?

References

No external references are newly cited in this cycles 7-9 report. Source references accumulated by ingestion and enrichment agents remain in REFERENCES.md.

Appendix: Implementation Details

Session Inventory

	Cycle	Source ID	Role	Contents
	7	00ecd932-4314-43ed-8a94-b64661660769	Researcher	Barrier 1 canonical-member repair brief
	7	e63dca56-ad6d-4877-9e8c-6cf590e38f5e	Worker	Repaired Barrier 1 scripts, regenerated substrate outputs, ran validators
	7	67171417-7a93-4bef-bcf8-16b65a264c6d	Auditor	Validated Barrier 1 repair
	8	3edad7dc-2222-4bec-8bc0-2739c0c80d5b	Researcher	Wave 2 six-track fan-out brief
	9	7e8a8f0a-786b-484b-8c69-d41b89087cc2	Worker	Barrier 2 post-merge integration and conformance validation

Code and Artifact Inventory

File	Lines	Purpose
tools/validate_barrier2_track_enrichment.py	247	Cross-track Barrier 2 enrichment conformance validator
reports/barrier2_wave2_integration_report.md	82	Human integration report for six Wave 2 branches
tracks/track1/docs/ENRICHMENT_AUDIT.md	251	Track 1 reticulation enrichment audit
tracks/track2/docs/ENRICHMENT_AUDIT.md	130	Track 2 ghost-partner seed enrichment audit
tracks/track3/docs/ENRICHMENT_AUDIT.md	269	Track 3 convergence trait-membership audit
tracks/track4/docs/ENRICHMENT_AUDIT.md	57	Track 4 domestication enrichment audit

File	Lines	Purpose
tracks/track5/docs/ENRICHMENT_AUDIT.md	297	Track 5 chemodiversity enrichment audit
tracks/track6/docs/ENRICHMENT_AUDIT.md	72	Track 6 offline probe construction audit
data/barrier2_track_enrichment_conformance.json	n/a	Machine-readable PASS summary

Test Results

Cycle 7 Barrier 1 repair:

```
PASS: Barrier 1 substrate validation (363237 nodes, 641183 retained hyperedges)
10 passed in 32.16s
promise_check: exit 0
org_check: exit 0, warnings only
```

Cycle 9 Barrier 2 integration:

```
PASS: Barrier 2 track enrichment conformance (6 tracks checked)
41 passed in 20.87s
PASS: Barrier 1 substrate validation (363237 nodes, 641183 retained hyperedges)
promise_check: exit 0
org_check: exit 0, warnings only
```

Cross-Reference Map

Origin	Consumer	Carried-forward value
Barrier 1 repaired substrate	All Wave 2 branches	Read-only accepted-key and canonical-member namespace
Track 1 reticulation enrichment	Barrier 2 integration	ready_data_limited; 25 pending-crosswalk rows
Track 2 ghost-partner enrichment	Barrier 2 integration	Seed evidence only; zero prediction rows
Track 3 convergence enrichment	Barrier 2 integration	209,297 memberships; no convergence_signature
Track 4 domestication enrichment	Barrier 2 integration	6 observed edges; climate and CWR shortfalls
Track 5 chemodiversity enrichment	Barrier 2 integration	Dr. Duke dominance and source-density diagnostics
Track 6 offline probe	Barrier 2 integration	210 offline questions; no live provider responses
prediction_ledger.tsv	Barrier 2 validator	Confirmed empty during enrichment integration

Remaining Gaps

Track 1 requires accepted-key recovery or a raw-name-aware instrument strategy before canonical polyploid validation can be used. Track 4 requires CWR expansion and observed

bioclim extraction before climate substitution predictions. Track 5 requires non-Duke counterfactual coverage before chemodiversity prediction can be separated from screening intensity. Track 6 requires a future free/open/local execution path before model scoring exists. Track 3 requires Barrier 2 agreement on how to treat the large _other bucket.